

JeevaFlux - 8-Month Research & Development Roadmap

12-week prototype-to-pilot preparation plan

JeevaFlux • Team SYMBIONICS • Smart India Hackathon 2026 • SIH26181

1. Roadmap Objective

The 12-week roadmap converts the JeevaFlux concept into a measurable, testable prototype by prioritizing sensing, edge intelligence, sensor fusion, personalization, mobile integration, offline resilience and validation.

2. Three-Month Overview

Milestone	Month 1	Month 2	Month 3
Requirements & architecture	✓		
Sensor integration & data pipeline	✓	✓	
Signal quality / preprocessing	✓	✓	
Edge AI / anomaly models	✓	✓	✓
Environmental-risk fusion		✓	✓
Personalized baseline		✓	✓
Mobile companion		✓	✓
Offline + synchronization		✓	✓
Validation & pilot protocol			✓
Documentation & presentation			✓

3. Month 1 — Foundation & Sensing

Weeks	Milestones	Expected Outputs
1	Requirements, user scenarios, architecture	SRS draft, system architecture, risk-module definitions
2	Sensor selection/integration	Working sensor acquisition pipeline
3	Signal quality & preprocessing	Filtering/calibration routines, quality flags
4	Baseline data collection	Initial dataset structure, baseline-generation method

4. Month 2 — Intelligence & Integration

Weeks	Milestones	Expected Outputs
5	Feature engineering	Windowed physiological/environmental features
6	Edge anomaly/risk inference	Lightweight model or rule+ML prototype

7	Multimodal sensor fusion	Risk-scoring pipeline and test scenarios
8	Personalization + mobile integration	Baseline adaptation, alert UI, event history

5. Month 3 — Resilience, Validation & Pilot

Weeks	Milestones	Expected Outputs
9	Offline-first implementation	Local storage, offline alert workflow
10	Synchronization/security	Store-and-forward sync, consent/access controls
11	Controlled validation	Accuracy, false-alert, latency and power results
12	Pilot preparation & documentation	Pilot protocol, final report, demo and future-work plan

6. Research Work Packages

A. Physiological sensing

- Assess heart rate, SpO₂, skin temperature, motion/activity, sleep/recovery and GSR inputs where implemented.
- Characterize noise, missing data and motion artifacts.

B. Environmental intelligence

- Acquire temperature/humidity and particulate/air-quality inputs.
- Define exposure windows and context features.

C. Edge AI

- Evaluate lightweight models suitable for MCU/edge execution.
- Benchmark latency, memory and energy consumption.

D. Sensor fusion

- Compare single-sensor and multimodal risk decisions.
- Develop quality-aware fusion to reduce false alarms.

E. Personalization

- Establish user baseline from normal periods.
- Evaluate adaptation without allowing unsafe threshold drift.

F. Mobile/offline

- Implement alert UX, local event storage and recovery after connectivity loss.
- Test synchronization integrity.

G. Validation

- Define labeled scenarios and ground truth procedures.
- Measure sensitivity, specificity, false alerts, warning lead time, power and usability.

7. Deliverables by End of 12 Weeks

- Integrated sensing prototype.

- Edge preprocessing and inference pipeline.
- Multimodal risk engine with documented assumptions.
- Personalized baseline prototype.
- Mobile companion prototype.
- Offline/store-and-forward workflow.
- Controlled validation report with measured KPIs.
- Pilot protocol, technical documentation and presentation/demo package.

8. Decision Gates

Gate	Proceed When
Gate 1 — Sensing	Measurements are stable enough for controlled testing and quality checks are implemented.
Gate 2 — Intelligence	Risk module produces repeatable outputs on labeled scenarios.
Gate 3 — Integration	Device, edge inference and mobile alert flow work end-to-end.
Gate 4 — Offline	Warnings remain functional without network and records recover correctly.
Gate 5 — Pilot	Safety limitations, privacy process, validation metrics and support workflow are documented.

